Public Google Docs MCP Server Spec

A lightweight development spec for exposing public course documents through a Streamable HTTP MCP server.

Goal

Build a lightweight Streamable HTTP MCP server that exposes a curated set of public Google Docs to participants. The server does not authenticate users and does not ingest or process document contents in the MVP. Participant-side clients or harnesses fetch and process the documents directly.

Assumptions

- All Google Docs are public or otherwise accessible without participant authentication.
- The participant harness can fetch public HTTPS URLs.
- The MCP server is read-only.
- The first version prioritizes simplicity over search, indexing, or server-side retrieval.

MCP Transport

- Expose one Streamable HTTP MCP endpoint, conventionally **/mcp**.
- Support MCP resource discovery.
- Require no auth for the MCP endpoint.
- Expose no write tools.

Resource Model

Each Google Doc is exposed as an MCP resource with these fields:

- **uri**: direct Google Docs export URL
- **name**: stable machine-readable identifier
- **title**: human-readable document title
- **description**: short explanation of when to use the document
- **mimeType**: preferably text/markdown or text/plain

Preferred Google Docs URL format:

```
https://docs.google.com/document/d/{DOC_ID}/export?format=md
```

Fallback format:

```
https://docs.google.com/document/d/{DOC_ID}/export?format=txt
```

Avoid ordinary /edit or /view links as the primary resource URI because those return the Google Docs web app rather than clean document text.

MVP Behavior

The server implements resource listing through **resources/list**.

Example resources:

```
[
  {
    "uri": "https://docs.google.com/document/d/FAQ_DOC_ID/export?format=md",
    "name": "onboarding-faq",
    "title": "Onboarding FAQ",
    "description": "Answers common participant setup and onboarding questions.",
    "mimeType": "text/markdown"
  },
  {
    "uri": "https://docs.google.com/document/d/PROJECT_1_DOC_ID/export?format=md",
    "name": "project-1-handout",
    "title": "Project 1 Handout",
    "description": "Instructions and requirements for Project 1.",
    "mimeType": "text/markdown"
  }
]
```

Participant Harness Behavior

- Connect to the MCP server.
- Call resources/list.
- Read the returned resources.
- Fetch each resource URI directly over HTTPS.
- Chunk, embed, summarize, search, or otherwise process the document locally.
- Preserve source attribution using the resource title and URI.

Optional Compatibility Extension

Some MCP clients may not automatically fetch external HTTPS resources. To support those clients, add **resources/read**.

In that version, resource URIs may remain direct HTTPS export links, or become stable custom URIs like:

```
course-doc://onboarding-faq
course-doc://project-1-handout
```

If using custom URIs, resources/read maps each custom URI to the corresponding Google Docs export URL, fetches the content server-side, and returns the text.

Recommended stages:

- Version 1: direct HTTPS resources only.
- Version 2: add resources/read while keeping stable resource names.
- Version 3: optionally add search tools if participant clients need server-side retrieval.

Non-Goals

- No participant authentication.
- No Google OAuth.
- No document editing.
- No private Google Drive access.
- No server-side vector database in the MVP.
- No answer-generation tool in the MVP.
- No deployment-specific requirements.

Security Notes

- Treat document contents as untrusted input from the client/model perspective, even if authored internally.
- Keep resource descriptions factual and minimal.
- Do not expose hidden or private docs by accident.
- Prefer a hardcoded allowlist of document IDs over accepting arbitrary Google Docs URLs from users.

Suggested Implementation Shape

Use a small MCP server with a static document registry.

```
const docs = [
  {
    name: "onboarding-faq",
    title: "Onboarding FAQ",
    description: "Answers common participant setup and onboarding questions.",
    uri: "https://docs.google.com/document/d/FAQ_DOC_ID/export?format=md",
    mimeType: "text/markdown"
  },
  {
    name: "project-1-handout",
    title: "Project 1 Handout",
    description: "Instructions and requirements for Project 1.",
    uri: "https://docs.google.com/document/d/PROJECT_1_DOC_ID/export?format=md",
    mimeType: "text/markdown"
  }
];
```

The MCP server should expose this registry through resources/list.

References

- MCP Resources: https://modelcontextprotocol.io/specification/2026-07-28/server/resources
- MCP Streamable HTTP: https://modelcontextprotocol.io/specification/2026-07-28/basic/transports/streamable-http
- Google Docs export formats: https://developers.google.com/workspace/drive/api/guides/ref-export-formats